Exp No: 2

Date: 23/7/26 Study of Network Cables

Aim: Study of different types of network cables.

a) Understand different types of network cables.

Different types of cables used in networking:

1) Unshielded twisted pair (UTP) cable.
2) shielded twisted pair (STP) Cable
3) Coaxial Cable.
4) Fibre optic cable.

Cable Type	category	maximum Data Transmission	Advantages / Disadvantages	Application / use
UTP	category 3	10 bps	Adv: *cheaper in cost * Easy to install as they have a smaller overall diameter Dis: *more prone to (Emi) Electromagnetic interference and noise	10 Base-T Ethernet
	Category 5	upto 100 mbps		Fast Ethernet, Gigabit Ethernet
	Category 5e	1 Gbps		Fast Ethernet Gigabit Ethernet
STP	category 6, 6a	10 Gbps	Adv: * shielded * Faster than UTP * Less susceptible to noise and interference Dis: * Expensive * Greater installation effort.	Gigabit Ethernet, 10 G Ethernet (55m) widely used in data centers.
SSTP	category 7	10 Gbps		Gigabit Ethernet, 10 G Ethernet (100 m)

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coxial cable	RG-6 RG-59 RG-11	10 -100 mbps	Adv:- * High bandwith * immune to interfaca * Low Loss bandwith * versatile Dis:- *Limited distance * cost * size is bulky	speed of signal is 500 m Television network High speed internet connections.
Fibre optics Cable	single mode multi mode	100 GbPS	Adv:- * High speed * High bandwith * High security * Long distance Dis:- * Expensive * Requires skilled installers.	* maximum distance of fibre optics cable is around 100 meters.

Result:-

Static IP address were successfully assigned to computers. connectively was verified using ping commands between hosts on the same subnet.

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